This project is the first systematic attempt to study the distance distribution of Reed-Muller codes across all parameters, including large finite fields. Its practical applications are wide-ranging, most importantly in cryptography (identity systems, blockchain scalability) wherein distance distribution of codes helps determine the soundness of zero-knowledge proofs, in quantum computing to determine quantum code's error-correction capability, and in storage system design to extend flash memory lifetime. These innovations strive to make our computing infrastructure more secure, reliable, and efficient. In this project, I answer a fundamental and long-standing open question in coding theory called the distance distribution problem (or coset weight distribution problem), which asks how distances between words and codewords are distributed across the space of all possible messages. Although special cases such as weight distribution of Reed-Muller codes or distance distribution of Reed-Solomon codes have been studied in this regard, the general problem for Reed-Muller codes over arbitrary finite fields stands unsolved due to the complex nature of multivariate polynomials. I resolve this nearly 50-year-old open problem by giving the first general error bounds on the distance distribution of Reed-Muller codes. This is algebraically equivalent to counting multivariate polynomials over finite fields with prescribed degree, coefficients, and number of zeroes. The unique approach I take analyzes character sums over multivariate polynomial quotient rings over a finite field, which have not been studied in more than one variable prior to this work. As a corollary of this work, I derived error bounds for the asymptotics of the distance distribution.